UNIT- II

Ideation Process

Clear Articulation of problem statement with focus on latent needs; Brainstorming potential solutions; Ideation methods with case-study based approach to using Systematic Inventive Thinking (SIT) Methods such as Addition, Subtraction, Multiplication, Division and Task Unification Strategic Innovation for competition in future: Linear Innovation vs. non-linear innovation, Understanding and identifying weak signals, 3-box thinking, 3-Box framework and Box-3 ideation

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Types of Needs

- **Choose** a problem area and conduct empathy process using any of the empathy tools or a combination of tools.
- **Next step** is to define Need statements which forms the **DEFINE** state of the Design process,
- **Need statements** are the insights that you have gathered during empathy.
- Need statements can be classified as **Primary, Secondary and Latent Needs**.

- **Primary Needs:** Fundamental set of needs
- **Secondary Needs:** *Necessarily* connected to one or more primary needs
- **Latent Needs:** Unexpressed needs, typically identified through observation
  - **Difficult to express**
  - **Even users may not know their own needs!**
  - **Identifiable only through empathy: Addressing them brings great user satisfaction/ customer delight! — "Innovation"**

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Primary Needs

Here are some questions to ask to determine your primary customer's needs:

- What do we believe our primary customers value?
- What creates customer happiness?
- What delights them?
- What are their needs, not wants?
- What problem can your offering solve?
- What pain can you eliminate?
- What are their risk factors?
- Are they in your primary sector?

Secondary Needs

Here are some questions to get you thinking about your secondary customer:

- Who else must be satisfied?
- Who purchases the product?
- Are they the ultimate decision maker?
- Who else plays a role in the decision-making process?
- What are their needs?
- When is it important to focus on the secondary customer?
- Why is it important to understand the secondary customer?

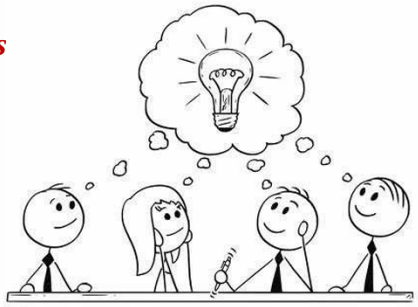
Addressing Latent Needs

- ***Address Unmet customer Needs – 'bring unexpected delight***
- ***Help understand emotionality of users/customers***
- ***Unless latent needs are known, you only think that you know***
- ***You know your customers, only if you know their latent needs***
- ***"You've got to start with the customer experience and work backwards to the technology. You cannot start with the technology and try to figure out where you are going to sell it" – Steve Jobs***

Brainstorming

Brainstorming is a part of design thinking used in the ideation phase.

Brainstorming is a method design teams use to ***generate ideas*** to solve clearly defined design problems.



Design teams gather for a brainstorming session in a room with a large board/wall for pictures/Sticky Notes.

A ***good mix of participants*** will expand the experience pool and therefore broaden the idea space.

Brainstorming → Brainstorming Rules

1. Set a time limit

2. Begin with a target problem/brief

3. Refrain from judgment/criticism

4. Encourage weird and wacky ideas

5. Aim for quantity

6. Build on others" ideas

7. Stay visual

8. Allow one conversation at a time

Brainstorming

The following steps steer the brainstorming process towards producing ideas that can be developed further.

Visualisation

Quick ***visual aids*** can be used to aid brainstorming and focus attention

Groups and voting

The brainstorming participants can be split into ***smaller groups***, each of which is ***tasked with finding solutions*** to particular concerns or challenges.

Scribble, say, slap



Brainstorming

Assessment criteria

Finally, to ***assess the ideas generated*** requires the need for ***criteria*** against which they can be assessed. Criteria can include

- ***Cost,***
- ***Resources required,***
- ***Necessary resources available,***
- ***Time factors etc..***

ARTICULATION OF PROBLEM STATEMET (Ideation)

IDEATION

"Ideation is the mode of the design process in which you concentrate on idea generation. Mentally it represents a process of "going wide" in terms of concepts and outcomes. Ideation provides both the fuel and also the source material for building prototypes and getting innovative solutions into the hands of your users."

– d.school, An Introduction to Design Thinking PROCESS GUIDE

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Ideation Techniques

- *Brain Storming*
- *Mind Maps*
- *Mashup*
- *Analogues – 6Hats*
- *SCAMPER (Substitute, Replace, Elimination)*
- *Systematic Inventive Technique*

(Subtraction, Division, Multiplication, Task Unification, Attribute)

- *3 Box Thinking*



Principles

- Closed World – Thinking inside the box
- Qualitative Change
- Function Follows Form
- Form Follows Function
- Path of Most Resistance
- Cognitive Fixedness – Functional, Structural
- Near Far Sweet

Systematic Inventive Thinking

The four key aspects of SIT.

Inside the Box

FFF Process

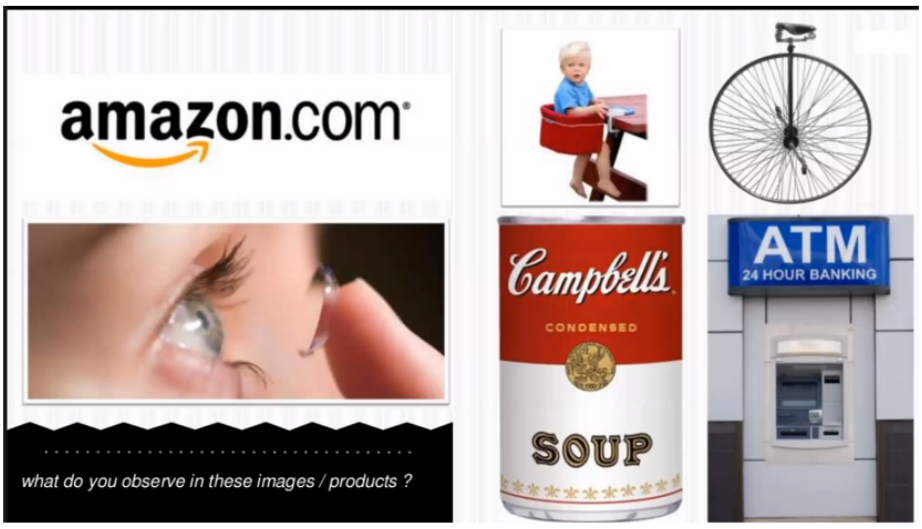
Five Innovation Patterns

Breaking Fixedness

SIT - a reliable methodology based on profound research - leads you through a process of provocation, challenging your existing assumptions, with the goal to create ideas that surprise you.



5 principles of SIT are in below examples:



SUBSTRACTION:

1. Amazon ***avoided*** people roaming out
2. lens ***removed*** frames
3. Chair ***avoided*** legs
4. Cycle ***removed*** chain and handle, one tyre
5. Soups ***removed*** water (concentrated)
6. ATM ***removed*** the usage of Bank



Subtraction	Multiplication
	
The elimination of core components rather than an addition of new systems and functions – „The path of most resistance“.	A multiplication (or addition) of elements already existing in the product along with a required adjustment – a qualitative change.

Multiplication







Subtraction



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Division



The division of a product and/or its components according to function or symmetry, thus adding degrees of freedom.





Subtraction	Multiplication	Division	Attribute Dependency
The elimination of core components rather than an addition of new systems and functions – „The path of most resistance“.	A multiplication (or addition) of elements already existing in the product along with a required adjustment – a qualitative change.	The division of a product and/or its components according to function or symmetry, thus adding degrees of freedom.	The creation/removal or symmetries or dependencies between existing product properties (e.g. color changes with temperature, etc.).

Attribute Dependency





Task Unification





Task unification



- **Linear Innovation**

Linear Innovation

What it is:
A step-by-step, sequential approach where each phase follows the previous one in a fixed order.

Typical flow:
Research → Ideation → Design → Development → Testing → Launch

Key characteristics:

- Predictable and structured
- Clear milestones and timelines
- Assumes the problem is well-defined from the start
- Changes late in the process are costly

Where it works best:

- Incremental improvements (e.g., upgrading an existing product)
- Stable markets with known user needs
- Engineering-heavy or compliance-driven projects

Limitations in design thinking:

- Little room to revisit assumptions
- User insights discovered late may not influence core decisions
- Innovation tends to be incremental, not breakthrough

Non-Linear Innovation (Design Thinking style)

What it is:
An iterative, flexible approach where teams move back and forth between stages based on learning and feedback.

Typical flow:
Empathize ↔ Define ↔ Ideate ↔ Prototype ↔ Test
(…and then loop again)

Key characteristics:

- Human-centered and exploratory
- Embraces ambiguity and uncertainty
- Learning-driven, not plan-driven
- Failure is used as feedback, early and often

Where it works best:

- Complex or "wicked" problems
- New products, services, or experiences
- Situations with unclear or evolving user needs

Strengths in design thinking:

- Deeper user empathy
- Higher chance of disruptive innovation
- Faster learning through rapid prototyping
- Solutions evolve *with* users, not just *for* them

Aspect	Linear Innovation	Non-Linear Innovation
Process	Sequential	Iterative & looping
Problem definition	Fixed early	Continuously reframed
User involvement	Limited, often early	Continuous
Risk handling	Avoided	Embraced as learning
Innovation type	Incremental	Breakthrough / radical

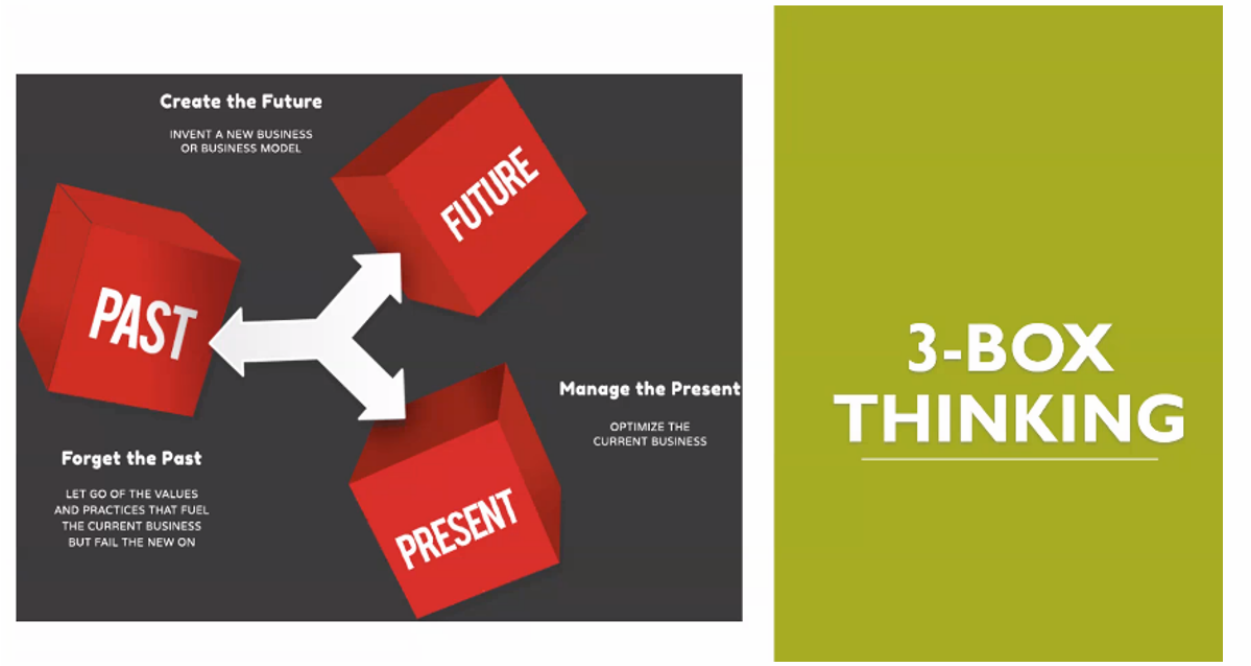
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- **Understanding and identifying weak signals,**

Design thinking is about **anticipating needs, not just reacting to them**. Weak signals help designers:

- Spot **future user needs** before users can articulate them
- Avoid incremental improvements and instead enable **breakthrough innovation**
- Design for **what's coming**, not only what exists now
- Reduce risk by exploring uncertainty early

Many major innovations started as weak signals that were ignored:

- Touchscreens as "finger smudging problems"
- Emojis as "unprofessional"
- Ride-sharing as "unsafe and niche"



A *mindset* for balancing present performance, past habits, and future innovation.

3-Box Thinking = *how you think*

3-Box Framework = *how you manage*

Box-3 Ideation = *how you invent the future*

The three boxes (popularized by Vijay Govindarajan):

1. **Box 1 – Manage the Present**
   - Optimize current operations
   - Improve efficiency, execution, KPIs
2. **Box 2 – Selectively Forget the Past**
   - Identify practices, beliefs, or assumptions that no longer work
   - Stop defending legacy thinking
3. **Box 3 – Create the Future**
   - Experiment, innovate, explore new models
   - Invest in ideas that don't yet have clear ROI

3 - Box Framework

Structured model or *management framework* that formalizes 3-Box Thinking.

Think of this as the **operational version** of the mindset.

How it's applied:

- Strategy planning
- Leadership development
- Portfolio allocation
- Innovation governance

Typical questions per box:

- **Box 1:** What must we execute better *now*?
- **Box 2:** What should we stop doing or unlearn?
- **Box 3:** What experiments are we funding for the future?

Box-3 Ideation

A *specific innovation activity* focused only on **future-oriented ideas**.

This is **not** about improving today—it's about imagining *what could exist*.

Characteristics:

- Long-term (3–10+ years)
- High uncertainty
- Often disruptive or non-obvious
- Separate from core business constraints

Example prompts:

- "If we were starting from scratch today, how would we solve this?"
- "What customer problem will exist in 10 years that doesn't exist now?"
- "What would kill our current business—and should we build it ourselves?"